

GEORGE LARSON

Email: george.g.larson@gmail.com Phone: 412-945-0101 Location: Pittsburgh, PA Portfolio: <https://georgelarson.me> Contributions: <https://georgelarson.me/contributions> GitHub: <https://github.com/georgeglarson> LinkedIn: <https://www.linkedin.com/in/georgelarson>

SUMMARY

Updated 2026-07 Director of Technology with 25+ years building and operating software, infrastructure, and manufacturing systems. I have led teams at Sony Electronics, scaled a medical billing startup to national reach, programmed PLCs on production lines, and directed technology for a mid-size manufacturer. I do Linux platforms, DevOps automation, ecommerce, and security programs. I stay close to the keyboard even when the job is strategic. I build AI tooling that actually ships: multi-agent infrastructure, MCP servers, model-routing pipelines, and a personal review methodology now shipping surgical fixes upstream in mitmproxy, n8n, ComfyUI, and Charm's Crush.

CORE STRENGTHS

Software Development and Engineering

- Full stack delivery across PHP, Perl, SQL, Python, Go, Rust, C#, TypeScript, C++. PHP and Perl have been the career staples; the rest sit alongside.
- Code review, automated testing, and performance tuning for legacy and greenfield systems.

DevOps and Infrastructure Automation

- CI and CD design, infrastructure as code, container orchestration, and repeatable releases on self-hosted runners.
- Observability, incident response, high-availability design (replication, backups, CDN), and pragmatic SLO tracking for hybrid environments.

Technology Leadership and Strategy

- Strategic roadmaps that connect executive goals to team-level delivery.
- Calm stewardship for blended internal, offshore, and vendor teams.

Security and Privacy

- Network hardening, IAM, HostedScan/ZAP/Nikto assessment loops, and zero-trust rollouts.
- Regulated-data guardrails and tabletop response drills.

Data Engineering and Automation

- OCR digitization, ETL pipelines, semantic search, and knowledge-base design.
- Workflow automation using scripting, open models, and RAG pipelines where the payoff is real.

OPEN SOURCE PROJECTS

Recent Upstream Contributions Merged

- kputnam/stupidedi #288, #289: re-enabled the X12 4010 PO850, AR943, and RE944 EDI transaction sets, whose autoloader had been off after the segment definitions outgrew their element lists. Merged July 2026.
- juspai/neurolink #1134: moved system messages to the AI SDK generateText system option; shipped in v9.82.0 on npm. Merged July 2026.
- ferriskey/ferriskey #1133: added the RFC 8693 TokenExchange grant type and its domain value objects to the Rust auth server. Merged July 2026.
- OpenHands/agent-canvas #1368: parent-grouped workspace-folder dropdown so nested paths stay organized. Merged July 2026.
- OpenHands/agent-canvas #1434: acp-docker example inherits its agent-server image from config/defaults.json instead of pinning below the Canvas compatibility floor. Merged June 2026.
- mvanhorn/last30days-skill #564: pass(1) credential source, a Linux/Unix analog of the macOS Keychain integration. Merged June 2026.
- sst/models.dev #475: added Claude Opus 4.5 to the model catalog. Merged Dec 2025.
- sst/opencode #4748: documented the Venice.ai provider for the opencode coding agent. Merged Nov 2025.

In review

- livekit/agents #6280: duplicate same-target handoff tool call no longer reads as ambiguous and aborts the agent switch.
- allenai/open-instruct #1751: a -launcher flag so non-Ai2 users can run the training launcher locally without a Beaker cluster account.
- stryker-mutator/stryker-net #3694: extends the array-creation mutator to implicit array creation.
- mitmproxy/mitmproxy #8199, #8200: Flow.kill() resume-event and KillInjected fixes so teardown closes the client connection and disconnects in-transit flows.
- charmbracelet/crush #2749, #2750: LSP score-based client selection and single-file-server handling so single-file servers start without root markers.

Fixed upstream (idea shipped under someone else's commit)

- n8n-io/n8n #29516: per-runner consecutive-timeout counter for stuck task-runner channel detection and recovery; reduced detection latency from hours to ~6s. A maintainer re-implemented the idea on current master and credited the diagnosis by name (#35456); shipped in the 2.33 and 2.34 release lines.

deep-research-privacy (45 stars, TypeScript)

- Standalone privacy-first AI research tool: Brave Search, zero data collection, parallel queries with cited output, and an iterative search-and-synthesis pipeline.

Venice AI SDKs (2025 to 2026, archived)

- Published TypeScript and PHP SDKs (venice-dev-tools, venice-ai-php) for the Venice API. Dated craft work; repositories archived 2026.

LLM Showdown (2026)

- llm-showdown-data. Companion dataset for the blog series benchmarking 15+ free coding models on a \$25 VPS using opencode and a ralph-wiggum-style iterative loop.
- Designed and runs the evaluation rig; publishes per-round results comparing model behavior on real coding tasks.

Fracture (2 stars, TypeScript), Modernization Case Study (2026)

- Modernizes Mozilla's 2012 BrowserQuest into a contemporary multiplayer architecture with eight integrated AI services: RAG, semantic search, agent-driven triage, and content moderation.
- Live demo at fracture.georgelarson.me. The case study itself is the deliverable: how to take a frozen reference codebase from a major vendor and bring it forward without rewriting from scratch.

nullclaw + ironclaw, Agent Mesh (Zig + Rust, 2026)

- nullclaw: public-facing AI concierge ("Nully") on a hardened perimeter box. Tiered inference (Haiku for conversational triage, Sonnet for code

analysis). Ran a full self-hosted comms stack (Ergo IRC, gamja web client, Cloudflare-proxied WebSocket, Let's Encrypt TLS) before transitioning to a static-site origin.

- ironclaw: private personal agent on a separate system with distinct security boundaries; agent-to-agent routing ran between the two until the daemon stack was retired June 2026.
- Extended into a multi-node Agent Zero mesh with A2A routing, dedicated chat and utility models, and per-node project isolation.

netgrep (Rust), Creator and Maintainer (2026 to Present)

- Modern replacement for ngrep: live packet capture and pcap analysis with TCP stream reassembly, TLS 1.2/1.3 decryption, HTTP/2 and DNS parsing, and an interactive terminal UI.
- Maintains ngrep-compatible CLI flags; outputs colored text, JSON, hex dump, or pcap.

termshark (Go), Contributor and Maintainer (2026 to Present)

- Modernized fork of termshark, a terminal UI for Wireshark/tshark.
- Refactored architecture, improved test coverage, updated to Go 1.22+ idioms, and added an experimental browser-based web interface.

wiregraph (Rust, 2026 to Present)

- GPU-rendered network traffic visualizer. Hosts as nodes, flows as edges, real-time topology view of a live capture session.

WORK EXPERIENCE

Director of Technology, FM Expressions (2018 to April 2026)

- Led technology strategy for a mid-size manufacturing company serving ecommerce channels.
- Absorbed the infrastructure team and its full stack when the role was eliminated during a downturn, running technology and infrastructure together with no added headcount.
- Shipped production AI inside the business from 2023: a customer-facing DALL-E design assistant for custom heat transfers, an OpenAI/Pythagora code-generation tool on MongoDB, an AI knowledge platform (wiki.fmx.ink), a customer-service AI/BI integration, and a Python/FastAPI templating system bridging the Fanatics and VivX vendor pipelines.
- Improved database performance by 93 percent while driving critical systems to 99 percent uptime; cut MySQL slow-query volume by 96 percent to accelerate incident response.
- Built a knowledge management platform (Wiki.js, Gitea, Qdrant, ETL pipelines) that captures production logs, transcripts, and operator notes so teams can search, surface, and act on institutional knowledge in real time.
- Resolved conflicting manufacturing documentation, created a style guide and glossary, and instituted monthly and quarterly review cycles to keep safety and operations guidance consistent.
- Reduced a 1.03 GB legacy software repository (4,656 binary artifacts) to a clean, LFS-ready codebase, eliminating clone failures and enabling sustainable development practices.
- Hardened hybrid infrastructure with Authentik SSO, HashiCorp Vault, automated TLS renewal, firewall hardening (iptables, UFW, CrowdSec, fail2ban), DMARC/SPF/DKIM enforcement, HostedScan/OWASP ZAP sweeps across 36 public assets, and Red Sift OnDMARC (p=reject, 1.5M emails/month at 99.48% compliance) to keep critical/high findings at zero; closed 33 issues in the 60 days prior to separation.
- Ran the security program's people-and-email layer: Mimecast email security, a KnowBe4 security-awareness and phishing-simulation program, and live response to an active production compromise.
- Ran the collaboration and operations core: self-hosted GitLab CI/CD, Mattermost team comms (retired 2026), Redmine project tracking, a Zammad helpdesk clearing ~200 tickets/month with 68% closed inside 24 hours, and Uptime Kuma monitoring feeding alert channels.
- Maintained high-availability platforms across a heterogeneous Windows-and-Linux estate: a high-availability MySQL cluster and MSSQL, VMware ESXi with Datto backups, on-prem Active Directory with Entra ID hybrid identity (Group Policy, PowerShell), multi-vendor firewalls (Cisco, SonicWall, pfSense), Cisco Meraki switching, OpenVPN, and a staged CDN migration from self-hosted edges to Contabo and Cloudflare; led countless OS and database migrations.
- Built the Mashy internal dashboard plus tooling such as Psono vault, PrivateBin, phpMyAdmin/Adminer, DBeaver, knowledge mining utilities, and status pages so teams access credentials, data stores, and observability in one click.
- Drove adoption metrics for the knowledge stack to quantify time-to-answer gains for production and customer operations.
- Captured Meteor print-head PCAPs and tuned Nikto, Metasploit, and custom fuzz rigs to secure industrial protocols while leading the move from VPN toward zero-trust access.
- Implemented Agile delivery, ticketing, and analytics to align product, operations, and finance on throughput, defect, and margin targets.
- Built and operated integrations across the print and fulfillment vendor stack (Pace/EFI, Printronix, VivX, Fanatics, Alphabroder, SignTronics) plus ecommerce platforms (Shopify, Magento) and marketing tooling (HubSpot, Power BI), so the production floor, order pipeline, and analytics layer share a coherent data model.
- Administered the prepress production stack (Esko, Prinergy, RIP workflows) and 3PL logistics integrations that kept the manufacturing floor running.
- Built financial-modeling tools using AI/ML to analyze company performance and forecast scenarios; presented findings directly to executive leadership to inform strategic decisions during a difficult business period.

Senior Software Engineer and Manager, Universal SmartComp (2009 to 2016)

- Helped scale the company from a 50 person startup to a national leader in medical billing (Inc 5000, ranked #34 in Health and #503 overall).
- Designed the EDI framework, ACH transmission, ERP integration, payer portal, Gold Standard Review, and recovery analytics layers of the platform.
- Led internal and offshore teams across development and Linux infrastructure.
- Acted as security response lead, HIPAA/HITECH SME, and Linux SME for the medical billing platform; migrated project management to Redmine with custom plugins.
- Deployed VMware ESXi covering web, database, DHCP, DNS, and ERP servers across the platform.
- Built OCR bill entry validation that reduced errors and sped turnaround.

Software Engineer and Team Manager, Sony Electronics (2004 to 2009)

- Joined a service operation that was effectively bare shelves at start; directed software development, logistics, and technical operations for the Pittsburgh Customer Satisfaction Center from there.
- Designed and ran the TiVo firmware upgrade station for the United States market.
- Built hardware testing suites for CD drives, set-top boxes, PCBs, televisions, cameras, and MP3 players.
- Created an AS/400 parts ordering interface that saved millions in operational spend.
- Front-end software work delivered roughly 30 percent productivity gain and 12 percent expense reduction.
- Managed more than 50 combined technical and non-technical staff, including roughly 30 contracted personnel.
- Developed a VT-52 terminal emulation system used for inventory management.

Maintenance and Automation Technician, T. Marzetti Company (2000 to 2004)

- Led second shift maintenance for robotics, conveyors, and packaging machinery.
- Programmed PLCs and Adept V++ robotics in-house, tuned sensors, actuators, and industrial controls on the production line, and kept robotics development internal to limit vendor cost.

Earlier Career Highlights (1990s to 2000s)

- Software developer and webmaster for non-profit organizations.
- Engineer for The Sounding Board, a psychotherapeutic BBS in Columbus, Ohio; built custom C++ modules for Major BBS and kept one of the first internet-connected BBS systems alive.
- Contract developer for Daifuku USA working on material handling support systems.
- Paid programming and database work 1994 to 2000 via Pacific Judgment Recovery (Brice, OH), AGL Personnel (Grove City, OH), and a Remedy Staffing CNC contract.
- Two-time Business Professionals of America Chapter 13 president; state and regional wins in Computer Programming, Team Mathematics, and Extemporaneous Speech.

VOLUNTEER WORK

- Systems administrator for a volunteer Linux community, digital radio station, and open shell environment.
- Long-term support for animal shelters, including events, outreach, and facilities work.
- OCR and digitization project contributor for historical legal documents.

TECHNICAL SKILLS

Languages: PHP, Perl, SQL, Bash, Python, Go, Rust, Zig, C#, TypeScript, C++ Web and Data: MySQL, MSSQL, PostgreSQL, WordPress, Redmine, Gitea, Wiki.js Infrastructure: Linux, Active Directory, Entra ID, Group Policy, PowerShell, Apache, Nginx, Caddy, IIS, Kubernetes, VMware ESXi, Datto backups, Cisco firewalls, Cisco Meraki, SonicWall, pfSense, OpenVPN, CrowdSec, Let's Encrypt/ACME, Cloudflare CDN Cloud: AWS, Azure, Google Cloud, Proxmox Security and Identity: Authentik, HashiCorp Vault, Psono, CrowdSec, HostedScan, OWASP ZAP, Nikto, Metasploit, SSL/TLS, iptables, UFW, fail2ban, DMARC/SPF/DKIM, Mimecast, KnowBe4, DefectDojo, OpenVAS, zero-trust network design, incident response, CCPA/GDPR compliance work Collaboration: Agile, Scrum, Kanban, Redmine, Zammad, GitLab, CryptPad Mail Systems: iRedMail, Postfix, Dovecot Commerce: Shopify, custom ecommerce platforms, payment integrations Data and Automation: RAG design, OCR pipelines, semantic search, ETL, knowledge base design AI Agent Infrastructure: Agent Zero (A2A protocol, multi-node mesh, project isolation), Model Context Protocol (MCP server design), Composio toolkit integration, Venice AI SDK ecosystem (TypeScript/PHP SDKs), LiteLLM, multi-provider model routing across 12 authenticated providers (Anthropic, OpenAI, Z.AI GLM, MiniMax, Moonshot/Kimi, Airouter, Groq, Cerebras, Featherless, OpenRouter, NVIDIA, Mistral), tiered inference (utility/chat split), prompt caching, agent-driven CI and content pipelines, RAG and semantic search, agent security boundaries, scheduled autonomous tasks. Operational fluency across ten coding-agent CLIs (Claude Code, opencode, gemini-cli, Codex CLI, Kimi Code, OpenHands, Aider, Octofriend, Crush, letta-code) used as a benchmark and production stack. Systems and Networking: TCP/IP analysis, packet capture, PLC programming, industrial controls, prepress production (Esko, Prinergy, RIP workflows), IRC server administration Monitoring & Dashboards: Uptime Kuma, HostedScan, Red Sift OnDMARC, Dashy internal portal

EDUCATION AND PROFESSIONAL DEVELOPMENT

- Self-taught from C++ books as a teenager; built BBS modules in production before attending college.
- Continuous learner across hundreds of online courses, lectures, and certifications (see LinkedIn for details).
- Currently studying: Stanford CS336, Language Modeling from Scratch (2025).
- Additional coursework spanning systems engineering, network security, database internals, and cloud architecture through platforms including Coursera, MIT OpenCourseWare, and vendor certification programs.